

Table 1: Algorithm success rate on dataset collection 3 with 10.0% constraints

| Constraints Level<br>Method<br>Dataset | 0.1     |      |        |         |      |       |
|----------------------------------------|---------|------|--------|---------|------|-------|
|                                        | COPKM_P | KSKM | KSKM_E | BLPKMCC | PCCC | COPKM |
|                                        |         |      |        |         |      |       |
| synthetic_n1000_d2_k10                 | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 0.20  |
| synthetic_n1000_d2_k100                | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n1000_d2_k2                  | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n1000_d2_k20                 | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 0.70  |
| synthetic_n1000_d2_k5                  | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n1000_d2_k50                 | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n2000_d2_k10                 | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n2000_d2_k100                | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n2000_d2_k2                  | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n2000_d2_k20                 | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 0.00  |
| synthetic_n2000_d2_k5                  | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n2000_d2_k50                 | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n5000_d2_k10                 | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n5000_d2_k100                | 1.00    | 1.00 | 1.00   | 0.40    | 1.00 | 1.00  |
| synthetic_n5000_d2_k2                  | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n5000_d2_k20                 | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 0.60  |
| synthetic_n5000_d2_k5                  | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n5000_d2_k50                 | 1.00    | 1.00 | 1.00   | 0.90    | 1.00 | 0.00  |
| synthetic_n500_d2_k10                  | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 0.90  |
| synthetic_n500_d2_k100                 | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n500_d2_k2                   | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n500_d2_k20                  | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |
| synthetic_n500_d2_k5                   | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 0.50  |
| synthetic_n500_d2_k50                  | 1.00    | 1.00 | 1.00   | 1.00    | 1.00 | 1.00  |